ARYAMAN PURAV

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING **Brooklyn, NY**
Master of Science in Computer Science *Sept 2026 – May 2028*
Course work: Design and Analysis of Algorithms, Machine Learning, Foundation of Data Science

VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE **Mumbai, India**
Bachelor of Technology in Electronics and Telecommunication Engineering | GPA: 3.10 / 4 *Dec 2021 – May 2025*

EXPERIENCE

HACKIT TECHNOLOGY AND ADVISORY SERVICES **Mumbai, India**
Security Analyst – Red Team *Jun 2025 – Apr 2026*

- Designed and deployed an Android-based C2 framework via Google Firebase for covert communication; simulated endpoint evasion techniques including process hollowing and DLL injection to assess malware detection systems and refine red team strategy.
- Engineered a shell script and JavaScript tool to automate SMTP delivery and mailbox operations via SWAKS, MIME, and SOAP APIs, reducing manual setup time by ~80% and improving testing execution speed by ~70% across 50+ endpoints.

EATON INDIA INNOVATION CENTER (EIIC) **Pune, India**
Security Analyst Intern *Jun – Aug 2024*

- Built a Virtual OT Cybersecurity Lab using Factory I/O, OpenPLC, ScadaBR, and Kali Linux; emulated Modbus TCP attack scenarios achieving 100% threat containment and 92% accuracy in detecting unauthorized coil writes, while cutting setup costs by 75–90%.
- Uncovered undocumented UART and JTAG interfaces using JTAGulator; conducted structured penetration testing covering threat modeling, asset risk classification, and attack surface enumeration using Burp Suite, Metasploit, Wireshark, and Procmon — producing 2 formal security advisories.

CENTER OF EXCELLENCE IN COMPLEX AND NON-DYNAMICAL SYSTEMS **Mumbai, India**
Research Intern *Jun - Jul 2023*

- Conducted VAPT on 2 web applications and 5 Linux/Windows systems using Nmap, Metasploit, IDA Pro, and ExploitDB.
- Performed hardware reverse engineering via Ghidra, Binwalk, and UART interfaces to extract embedded device logic.

ACADEMIC PROJECTS & RESEARCH

Drone Security *May 2025*

- Identified MAVLink protocol weaknesses (command injection, heartbeat spoofing, telemetry eavesdropping) using Wireshark and Python; configured a SITL/Mission Planner testbed to execute real-time adversarial scenarios.
- Hijacked DJI Mavic Air control via WiFi de-auth attacks and SSID spoofing using Aircrack-ng; decoded 433 MHz / 5.8 GHz RF drone signals using RTL-SDR, HackRF, and GNU Radio.

Fuzzing Of Sumatra Pdf Reader *Dec 2024*

- Reverse-engineered libmupdf.dll using IDA, WinDBG, and x64dbg; developed a C++ DLL harness enabling WinAFL fuzz testing with DynamoRIO, improving code coverage by 35%+ and reducing input corpus size by 50%.
- Executed 5,000+ fuzzing iterations, tracing a heap use-after-free crash in pdf_new_document via WinDBG, demonstrating potential for critical memory corruption exploitation.
- Secured 1st place at **Smart India Hackathon 2024**.

Embedded Malware & Trojan Detection in Power Sector Devices *Dec 2023*

- Reverse-engineered firmware from 3 power-sector devices, identifying 7 critical vulnerabilities disclosed to CERT-In; built a Firmware Security Analyzer reducing analysis time by 70%
- Secured 1st place at **Smart India Hackathon 2023** and published findings at **ICCNS 2025**.

ACTIVITIES

Cybersecurity Lead, Google Developer Student Clubs, VJTI *Aug 2023 – Dec 2024*

- Led a 4-member team designing CTFs and delivering cybersecurity workshops to 40+ students.

Technical Executive, International Society of Automation, VJTI *Apr 2023 – Dec 2024*

- Managed technical operations and networking for ISA's annual symposium supporting 50+ attendees.

TECHNICAL SKILLS

- **Languages & Scripting:** Python, C, C++, Bash, JavaScript, Rust
- **Offensive Security:** Red Teaming, Penetration Testing (VAPT), Threat Modeling, Attack Surface Enumeration, Privilege Escalation, Lateral Movement, Post Exploitation, Credential Abuse, CSRF Exploitation, Android Testing
- **Malware & Evasion**: Malware Analysis, Static & Dynamic Analysis, Process Hollowing, DLL Injection, Sandbox Evasion, Memory Forensics, Fuzzing, Binary Exploitation, Reverse Engineering
- **Security Tools:** Burp Suite, Metasploit, Wireshark, IDA Pro, WinDBG, x64dbg, Ghidra, Binwalk, NetExec, Nmap, Procmon, DynamoRIO, Visual Studio, Android Studio
- **Platforms & Environments:** Linux (Kali, Ubuntu), Windows, Android, VMware / VirtualBox, Git, Flask, Django, Docker